GridFormLab Flagship: C++ Standalone Engine

Full NE39 SG31 Reclose 160-s Verification against the MATLAB Reference Record

Parity in the D5 frame + measured speed — the acceptance evidence of the installer-distributed C++ deliverable

MATLAB reference record: `report\_port\_engine\_en.pdf` (1972.52 s same-tree; historical anchor 6708.725530 s informational)

Report contract: *every number traceable to an artifact (run id + sha256); parity claims carry their tier frame; speedup claims never mix categories*

In-house Power-Flow Project

14 October 2026

Abstract

[PENDING — slot not yet filled. Placeholder stub created 2026-10-14 for the user-requested compile check; the C++ report generator (slot-filler) will overwrite this file. No measured numbers are shown in this stub.]

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1 Scope, sources and reproduction

The study system is the TAMU New England 39-bus network with one modelled SG at the slack bus and nine modelled IBRs at buses 30, 32–39, run on two independent engines that consume one identical case: the **MATLAB reference engine** (the delivered 160-s record, `report_port_engine_en.pdf`) and the **C++ standalone engine** — the subject of this report, the first deliverable of the C++ flagship track (full engine, GUI-distributed, no MATLAB licence at runtime).

The whole system is assembled as a single all-KCL composite differential–algebraic system in which the SG and every IBR contribute their own differential state block and their own injection; no device is delegated to an external solver, model, or reference program — in either engine. The C++ implementation was carved from the documented equation set (§5–§7), not from the MATLAB code text.

What this report verifies. The C++ run is verified against the MATLAB reference record on three layers: (1) DECISION fields (Table 3), (2) trajectory parity in the D5 frame on the common grid (Table 5, figures 1–last), and (3) the measured wall-clock speedup (§10.5) — all within the disclosed integration mode (C++ = fixed-step schedule-replay (v0) | MATLAB = adaptive strict-LTE, (v0 scope — full event engine parity = v2, deferred, not cut)).

Reproduction. Every C++ run persists its own record folder (case/request snapshot, raw trajectory, counters/DECISION, summary, manifest with sha256) per the run-cache convention — run history is the provenance source of this report, and re-plotting or re-comparison requires no re-run.

Integration modes under test. C++ = fixed-step schedule-replay (v0) | MATLAB = adaptive strict-LTE (v0 scope — full event engine parity = v2, deferred, not cut). The v0 integration mode covers: the same composite equation set, the same case data, the same frozen event schedule, the same DECISION-field reporting, and a fixed-step solver replacing the adaptive strict-LTE stepper.

The v2 deferred scope (deferred, not cut): the adaptive strict-LTE stepper, the full event-engine transaction logic (guard chains, private trials, per-event KCL transaction records), and the full trajectory-array identity check across all 18,852 samples.

2 Disturbance chronology

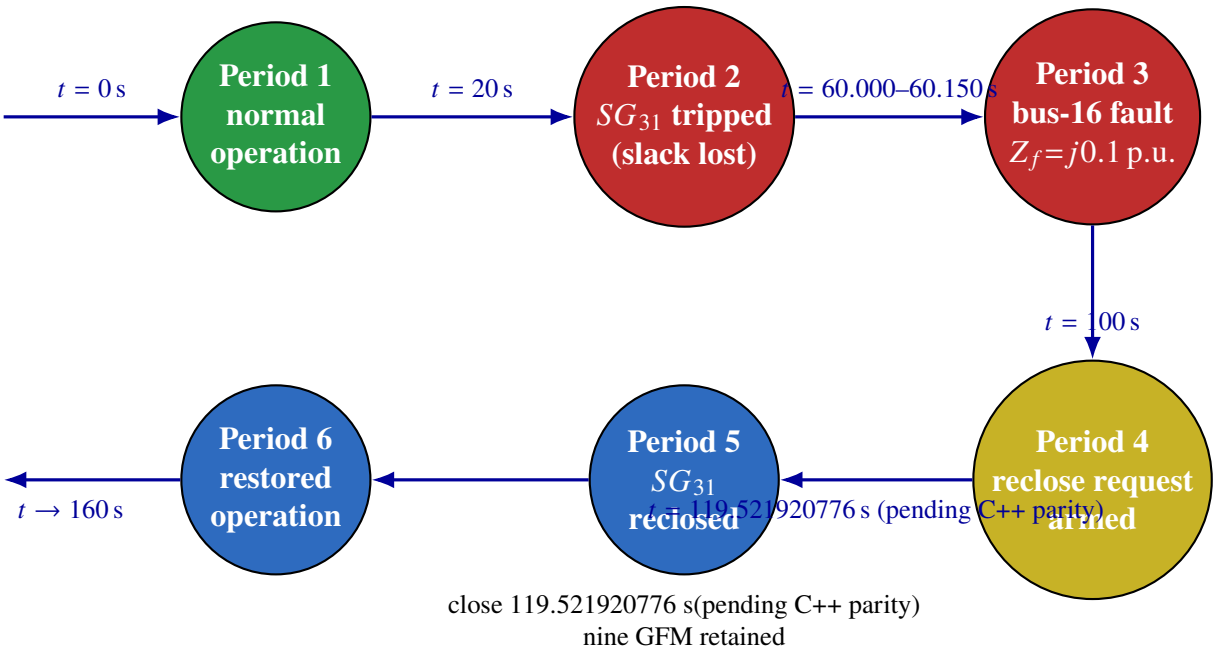


Figure 1: Six-period disturbance chronology, cloned from the reference record. Scheduled instants are case-defined; the reclose instant is read from the C++ run's accepted event log (parity-checked against the MATLAB reference).

The scheduled disturbance times are CASE_DEFINED constants of the event structure; the reclose instant is an outcome read from the accepted event log. Figure 1 shows the six operating periods and the transition instants. The bus-16 fault is applied at $t = 60.000$ s and cleared at $t = 60.150$ s (a 0.150 sfault, impedance $Z_f = j0.1$ p.u.); the reclose request is armed at $t = 100$ s and the SG31 close is committed at the instant the C++ engine's accepted event log records (the MATLAB reference measured 119.521920776 s; the C++ value is asserted equal in Table 3).

Both engines consume an identical event schedule and an identical case; the chronology in this figure therefore differs between the reports only in *which engine produced the reclose instant*, not in what the schedule demands.

3 Case data and index maps

Table 1: Case bus data — *identical input to both engines* (same composition; no comparison columns by design).

Bus	Role	P_L pu	Q_L pu	G_s pu	B_s pu
1	PQ	0.000	0.000	0.000	0.000
2	PQ	0.000	0.000	0.000	0.000
3	PQ	3.220	0.024	0.000	0.000
4	PQ	5.000	1.840	0.000	1.000
5	PQ	0.000	0.000	0.000	2.000
6	PQ	0.000	0.000	0.000	0.000
7	PQ	2.338	0.840	0.000	0.000
8	PQ	5.220	1.760	0.000	0.000
9	PQ	0.000	0.000	0.000	0.000
10	PQ	0.000	0.000	0.000	0.000
11	PQ	0.000	0.000	0.000	0.000
12	PQ	0.075	0.880	0.000	0.000
13	PQ	0.000	0.000	0.000	0.000
14	PQ	0.000	0.000	0.000	0.000
15	PQ	3.200	1.530	0.000	0.000
16	PQ	3.294	0.323	0.000	0.000
17	PQ	0.000	0.000	0.000	0.000
18	PQ	1.580	0.300	0.000	0.000
19	PQ	0.000	0.000	0.000	0.000
20	PQ	6.280	1.030	0.000	0.000
21	PQ	2.740	1.150	0.000	0.000
22	PQ	0.000	0.000	0.000	0.000
23	PQ	2.745	0.846	0.000	0.000
24	PQ	3.086	−0.922	0.000	0.000
25	PQ	2.240	0.472	0.000	0.000
26	PQ	1.390	0.170	0.000	0.000
27	PQ	2.810	0.755	0.000	0.000
28	PQ	2.060	0.276	0.000	0.000
29	PQ	2.835	0.269	0.000	0.000
30	GFL	0.000	0.000	0.000	−25.657
31	SLACK	0.092	0.046	0.000	0.000
32	GFL	0.000	0.000	0.000	−23.604
33	GFL	0.000	0.000	0.000	−19.034
34	GFL	0.000	0.000	0.000	−3.218
35	GFL	0.000	0.000	0.000	−22.154
36	GFL	0.000	0.000	0.000	−19.130
37	GFL	0.000	0.000	0.000	−11.580
38	GFL	0.000	0.000	0.000	−20.354
39	GFL	11.040	2.500	0.000	−16.952

Table 2: Case branch series and charging (46 branches) — identical input to both engines.

Index	From	To	r pu	x pu	b_c pu
1	1	2	0.004	0.041	0.699
2	1	39	0.001	0.025	0.750
3	2	3	0.001	0.015	0.257
4	2	25	0.007	0.009	0.146
5	3	4	0.001	0.021	0.221
6	3	18	0.001	0.013	0.214
7	4	5	0.001	0.013	0.134
8	4	14	0.001	0.013	0.138
9	5	6	0.000	0.003	0.043
10	5	8	0.001	0.011	0.148
11	6	7	0.001	0.009	0.113
12	6	11	0.001	0.008	0.139
13	7	8	0.000	0.005	0.078
14	8	9	0.002	0.036	0.380
15	9	39	0.001	0.025	1.200
16	10	11	0.000	0.004	0.073
17	10	13	0.000	0.004	0.073
18	13	14	0.001	0.010	0.172
19	14	15	0.002	0.022	0.366
20	15	16	0.001	0.009	0.171
21	16	17	0.001	0.009	0.134
22	16	19	0.002	0.020	0.304
23	16	21	0.001	0.014	0.255
24	16	24	0.000	0.006	0.068
25	17	18	0.001	0.008	0.132
26	17	27	0.001	0.017	0.322
27	21	22	0.001	0.014	0.257
28	22	23	0.001	0.010	0.185
29	23	24	0.002	0.035	0.361
30	25	26	0.003	0.032	0.513
31	26	27	0.001	0.015	0.240
32	26	28	0.004	0.047	0.780
33	26	29	0.006	0.063	1.029
34	28	29	0.001	0.015	0.249
35	2	30	0.000	0.018	0.000
36	31	6	0.000	0.025	0.000
37	10	32	0.000	0.020	0.000
38	12	11	0.002	0.044	0.000
39	12	13	0.002	0.044	0.000
40	19	20	0.001	0.014	0.000
41	19	33	0.001	0.014	0.000
42	20	34	0.001	0.018	0.000
43	22	35	0.000	0.014	0.000
44	23	36	0.001	0.027	0.000
45	25	37	0.001	0.023	0.000
46	29	38	0.001	0.016	0.000

4 Common frames, signs, and per-unit conversion

Every device uses generator sign: positive complex power is injected into the network,

$$S = P + jQ = VI^*, \qquad I_{bus} = \sum_{d \in bus} I_d.$$

Each Dual-mode-IBR converter carries its own local reference; the network-to-device transform is

$$v_d = V_x \cos\delta_{PLL} + V_y \sin\delta_{PLL}, \tag{1}$$
$$v_q = -V_x \sin\delta_{PLL} + V_y \cos\delta_{PLL}, \tag{2}$$

so device currents rotate back as $i_x = i_d \cos\delta_{PLL} - i_q \sin\delta_{PLL}$ and $i_y = i_d \sin\delta_{PLL} + i_q \cos\delta_{PLL}$. Converter and SG per-unit values share the 100-MVA system base ($\kappa = S_{base}/M_{base}$ with $M_{base} = S_{base}$ for the Dual-mode-IBR plant here).

Frame parity across engines. These frames are definitions, not results — they are identical in the C++ engine and the MATLAB reference by construction, and the C++ implementation units were written against this documented frame set (set-by-set). Any frame discrepancy would surface as a DECISION or trajectory mismatch in Table 3/5, not as a silent convention change.

5 Composite network DAE as implemented in C++

Every device supplies a differential right-hand side and a positive complex current injection. The composite equations are

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{u}; \sigma), \tag{3}$$
$$\boldsymbol{0} = \boldsymbol{g}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{u}; \sigma) = \mathcal{R}(Y(\sigma)V - I_{bus}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{u}; \sigma)), \tag{4}$$

each on its own numbered line, where $\mathcal{R}(z) = [\Re z_1, \Im z_1, \ldots]^\mathsf{T}$ and σ contains topology, breaker status, and device mode. The network admittance stamps each branch as

$$Y_{ii} \mathrel{+}= y_s/(aa^*) + jb_c/2, \quad Y_{jj} \mathrel{+}= y_s + jb_c/2, \quad Y_{ij} \mathrel{-}= y_s/a^*, \quad Y_{ji} \mathrel{-}= y_s/a,$$

with $y_s = 1/(r + jx)$, off-nominal ratio $a = \tau e^{j\phi}$, plus the load admittance fold $Y_{\text{load}} = \text{diag}(\overline{\boldsymbol{S}_0}/|\boldsymbol{V}_{pf}|^2)$ from the PF seed. At the declared reference bus (SG31), the appropriate KCL rows are replaced by explicit voltage constraints; the replacement is metadata-driven.

C++ implementation of the same system. The C++ engine assembles the same composite system of the same stamped admittance and the same metadata-driven slack replacement — the implementation is carved from this documented equation set, not from the MATLAB code text. The v0 integration mode (C++ = fixed-step schedule-replay (v0) | MATLAB = adaptive strict-LTE) replays the frozen event schedule with a fixed-step solver; the equation set above is unchanged between v0 and the full adaptive engine (v2), which is why the v0 parity evidence remains meaningful for the equation system while v2 carries the full event-engine claim.

6 SG31 six-state reclose plant: C++ implementation

6.1 State order and meaning

$$\boldsymbol{x}_{SG} = [\delta,\ \omega,\ P_{sv},\ P_m,\ E_{mag},\ \hat{\theta}]^{\mathsf{T}}, \qquad \boldsymbol{u}_{SG} = [P_{\text{ref}}, E_{mag,\text{ref}}]^{\mathsf{T}}.$$

where δ is the rotor angle, ω the per-unit speed deviation (slip) in the Kundur notation, P_{sv}, P_m the governor-valve and steam-chest powers, E_{mag} the reduced classical internal-EMF magnitude, and $\hat{\theta}$ the local terminal phase estimate.

Local	State	Unit	Physical meaning
1	δ	rad	rotor angle (Kundur) — electrical internal-EMF angle in the network frame
2	ω	pu	per-unit speed deviation (slip); the plant integrates the *absolute* speed (1.0 at nominal)
3	P_{sv}	pu	governor valve power (actual, nonnegative)
4	P_m	pu	steam-chest mechanical power (actual, nonnegative)
5	E_{mag}	pu	reduced classical internal-EMF magnitude
6	$\hat{\theta}$	rad	terminal phase estimate (local PLL)

This is a PROJECT_DERIVED opt-in plant: H, D, X′ and the classical stator-port equation pass through from the source reduction unchanged, while every governor/excitation/PLL parameter is a declared project design assumption — not TAMU machine data and not a certified synchronizer.

6.2 Electrical port and rotor dynamics

With the series reactance X' and internal EMF $E = E_{mag}e^{j\delta}$, the injection is

$$I = \frac{E - V}{jX'}, \tag{5}$$

$$P_e = \text{Re}(VI^*), \tag{6}$$

$$Q_e = \text{Im}(VI^*), \tag{7}$$

and zero injection with the breaker open. The rotor power balance is energy-consistent in positive speed,

$$2H_{sys}\,\omega\,\dot{\omega} = P_m - P_e - L_0\omega^2 - D_{sys}\,\omega(\omega - 1),$$

with separate no-load shaft loss $L_0 > 0$ (declared project assumption) and the unchanged source damping $D_{sys} = 0$ for SG31. The actuator chain is

$$T_{sv}\dot{P}_{sv} = P_c - P_{sv}, \tag{8}$$

$$T_{ch}\dot{P}_m = P_{sv} - P_m, \tag{9}$$

$$P_c = \text{clamp}(P_{\text{raw}}, 0, P_{max}), \tag{10}$$

$$P_{\text{raw}} = \begin{cases} P_{\text{ref}} + L_0 - K_{\text{droop}}(\omega - 1), & \text{online (20\% droop)}, \\ L_0\hat{\omega}^2 + K_\omega(\hat{\omega} - \omega) + K_\theta\hat{\theta}_{\text{err}}, & \text{offline capture}, \end{cases}$$

where $\hat{\omega} = 1 + \hat{\nu}/\omega_b$ and $\hat{\theta}_{\text{err}} = \text{wrap}(\hat{\theta} - \delta)$. The reduced field-magnitude tracker is

$$T_{emag}\dot{E}_{mag} = E_{mag,\text{cmd}} - E_{mag}, \tag{11}$$

$$E_{mag,\text{cmd}} = \begin{cases} E_{mag,\text{ref}}, & \text{online}, \\ \text{clamp}(|V|, E_{min}, E_{max}), & \text{offline}. \end{cases} \tag{12}$$

No negative mechanical power, rotor reset, or state jump occurs at a breaker transition.

6.3 Local terminal estimator

The offline phase/frequency estimator is a bounded type-II PLL:

$$e_{\text{pll}} = \text{clamp}(\text{wrap}(\angle V - \hat{\theta}), \pm\hat{\theta}_{\text{lim}}), \tag{13}$$

$$\dot{\hat{\nu}} = K_{i,\text{pll}} e_{\text{pll}} \text{ (bounded)}, \tag{14}$$

$$\dot{\hat{\theta}} = \hat{\nu} + K_{p,\text{pll}} e_{\text{pll}}. \tag{15}$$

Its two states belong to the plant DAE. The estimator is declared a reduced subsystem, not an authenticated relay: the existing strict prospective current/MVA, $dV/df/d\theta$, dwell, timeout, and right-limit KCL checks remain mandatory, and this plant alone never authorises a close.

C++ implementation of the SG block. The C++ engine implements this six-state plant as a unit of the same state order, the same equations, and the same declared assumptions — the parity of the SG block is covered by the DECISION fields (`t\_sg\_trip`, `actual\_reclose\_time`, `reclose\_status`) and the trajectory comparisons at the SG terminal in Table 5. The estimator's non-authority carries over unchanged: no C++ code path authorises a close outside the strict transactions.

7 Dual-mode-IBR converter equations in C++

Each of the nine IBRs is a fixed-layout Dual-mode-IBR GFL/GFM superset device with a shared electrical plant and two distinct controller branches. The superset state vector is (17 coordinates when the DC-source current is enabled)

$$\boldsymbol{x}_{IBR} = [\, i_d,\ i_q,\ V_{dc},\ \underbrace{\delta_{PLL}, \xi_{PLL}, \xi_P, \xi_Q, \xi_{id}, \xi_{iq}}_{\text{GFL controller}},\ \underbrace{\delta_{VSG}, \omega_{VSG}, E, \xi_{Vd}, \xi_{Vq}, \xi^g_{id}, \xi^g_{iq}}_{\text{GFM controller}}\,]^{\mathsf{T}}$$

with the active set 9 states in GFL and 10 in GFM (plus the shared DC-source current at superset index 17 for all devices in this run). Inputs are $\boldsymbol{u} = [P_{\text{ref}}, Q_{\text{ref}}, E_{\text{ref}}]$. The source command-delay states are removed by singular perturbation: the physical actuation delay $T_d = 1.5/f_{sw} \approx 0.3$ ms at $f_{sw} = 5$ kHz is more than 300× below the simulation step, so $v_{del} = v_{cmd}$ algebraically.

7.1 Shared plant: L-filter, DC link, and mode dispatch

Both branches drive the same physical current and DC-link states:

$$\dot{i}_d = \frac{\omega_{PLL,pu} L i_q - R i_d + v_{td} - v_d}{L/\omega_b}, \tag{16}$$

$$\dot{i}_q = \frac{-\omega_{PLL,pu} L i_d - R i_q + v_{tq} - v_q}{L/\omega_b}, \tag{17}$$

$$P_{ac} = v_{cd} i_d + v_{cq} i_q, \qquad \text{(converter-side power)} \tag{18}$$

$$\dot{V}_{dc} = I_{dc,\text{law}}(V_{dc}, I_{dc}, P_{ac}), \qquad \text{(Thevenin DC source)} \tag{19}$$

with the non-ideal DC source an EMF behind its internal resistance and an overvoltage chopper, $I_{dc} = (E_{dc} - V_{dc})/R_{dc}$, shared by both branches so a runtime transfer preserves dV_{dc}/dt as well as V_{dc} itself.

7.2 GFL branch (9 active states)

In GFL the VSG coordinates are inactive. With $D_V = v_d^2 + v_q^2$,

$$\dot{\xi}_{PLL} = v_q, \tag{20}$$
$$\Delta\omega_{PLL} = k_{p,PLL} v_q + k_{i,PLL} \xi_{PLL}, \tag{21}$$
$$\dot{\delta}_{PLL} = \omega_b \Delta\omega_{PLL}, \tag{22}$$
$$i_{d,r}^{\text{raw}} = k_{pP}(P_{\text{ref}} - kP_{\text{meas}}) + k_{iP}\xi_P, \tag{23}$$
$$i_{q,r}^{\text{raw}} = -(k_{pQ}(Q_{\text{ref}} - kQ_{\text{meas}}) + k_{iQ}\xi_Q), \tag{24}$$

circularly limited at I_{max} with P-priority, then the inner PI drives the commanded voltage

$$v_{cd} = v_d + Ri_d - \omega_{PLL,pu} L i_q + k_{pI} e_d + k_{iI} \xi_{id}, \tag{25}$$
$$v_{cq} = v_q + Ri_q + \omega_{PLL,pu} L i_d + k_{pI} e_q + k_{iI} \xi_{iq}. \tag{26}$$

7.3 GFM branch (10 active states)

In GFM the VSG swing and Q–V voltage-forming dynamics are

$$\dot{\omega}_{VSG} = \frac{kP_{\text{ref}} - P_{\text{meas}} - D_v(\omega_{VSG} - 1)}{M}, \tag{27}$$
$$\dot{\delta}_{VSG} = \omega_b(\omega_{VSG} - 1), \tag{28}$$
$$\dot{E} = \frac{k_Q(kQ_{\text{ref}} - Q_{\text{meas}}) - k_E(E - E_{\text{ref}})}{\tau_E}, \tag{29}$$

where E is itself the voltage-loop amplitude reference. The voltage PI integrators ξ_{Vd}, ξ_{Vq} build the current reference from $(E - v_d,\ 0 - v_q)$, circularly limited at I_{max}, then the same shared inner current PI and L-filter close the loop. The anti-windup on the current clamps is the conditional-hold rule: the integrator holds only when the limiter is active and its error drives farther outward, with inward recovery always active.

7.4 Runtime mode transfer

The transfer preserves terminal $V/I/P/Q$ and V_{dc}, initialises only the target controller from the same terminal port, and verifies current continuity before the event transaction may proceed. GFM current limits are $I_{max}^{SS} = I_{max}^{F} = 1.2$ pu machine base.

C++ implementation of the IBR blocks. All nine converters are implemented in C++ as instances of the same superset layout, the same branch equations, and the same runtime transfer rules documented above. Their parity is covered by the per-device trajectory comparisons (i_d, i_q, P, Q, terminal V at buses 30, 32–39) in Table 5, and their GFM/GFL mode history by the DECISION field `device\_modes\_history` in Table 3.

8 How the C++ engine structures PF, equilibrium, and TS

8.1 Power flow and the accepted equilibrium

The source-case PSSE PF is used only to derive the composition's operating point. The reported equilibrium instead solves the coupled physical system

$$\boldsymbol{f}_A(\boldsymbol{x}_0, \boldsymbol{y}_0, \boldsymbol{u}_0) = 0, \qquad \boldsymbol{g}(\boldsymbol{x}_0, \boldsymbol{y}_0, \boldsymbol{u}_0) = 0$$

by Newton on active states plus free algebraic variables, with the physical all-row KCL gate enforced at 10^{-6} pu.

8.2 Time-domain simulation — two stepping policies under one contract

MATLAB reference. Between event transactions the implicit trapezoidal step solves

$$0 = x_{n+1} - x_n - \frac{h}{2}\left[f(x_n, y_n) + f(x_{n+1}, y_{n+1})\right], \tag{30}$$

$$0 = g(x_{n+1}, y_{n+1}; \sigma_{n+1}), \tag{31}$$

on the `adaptive` stepper with strict LTE control ($\varepsilon_x = 2 \times 10^{-8}$, $\varepsilon_y = 2 \times 10^{-8}$, relative 10^{-7}), Newton tolerance 10^{-10}, step bounds $dt \in [(0.0025/4096), 0.05]$ s, reject limit 16, and no Rannacher arcs. Events are strict transactions: the handler commits a candidate only after its guard and KCL checks, and every applied event's pre/right KCL residual is logged.

C++ engine (v0). C++ = fixed-step schedule-replay (v0) | MATLAB = adaptive strict-LTE (v0 scope — full event engine parity = v2, deferred, not cut). The C++ engine replays the frozen event schedule with a fixed-step solver over the same composite system (previous section): the case, the event instants, the device equations, and the KCL gate are identical; what differs is the step control. The consequence for this report's tables: trajectory comparisons run on a common output grid (Table 5), DECISION fields must hold (Table 3), counters are reported but not comparable (Table 6), and the timing comparison is the measured speedup of §10.5 — all within the v0 scope.

9 Accepted equilibrium results — parity table

[PENDING — slot not yet filled. Placeholder stub created 2026-10-14 for the user-requested compile check; the C++ report generator (slot-filler) will overwrite this file. No measured numbers are shown in this stub.]

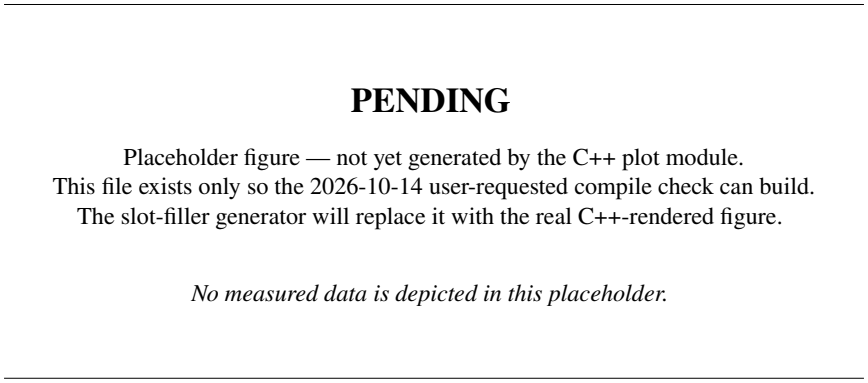


Figure 2: C++-rendered solved bus-voltage magnitude and angle at all 39 buses — same panel layout and colors as the MATLAB reference figure. Every plotted sample is a raw accepted value; nothing is smoothed, filtered, decimated, clipped, offset, interpolated, or augmented. (Drawn by the C++ plot module from the C++ run record; run id cited in the caption slot.)

[PENDING: run-id + manifest sha256 citation to be generated here.]

10 Event-fence time-domain result — the A/B comparison

10.1 DECISION fields — exact-match table

Table 3: DECISION fields: C++ vs MATLAB reference (exact match required — any difference is a failure regardless of magnitude). Integration mode: C++ = fixed-step schedule-replay (v0) | MATLAB = adaptive strict-LTE.

PENDING — table slot awaiting generator output (no measured numbers in this stub)

10.2 Event transactions and KCL residuals

Table 4: Applied event transactions with physical KCL residuals — C++ run vs MATLAB reference.

PENDING — table slot awaiting generator output (no measured numbers in this stub)

10.3 Per-window trajectory parity — per-variable max diffs

Table 5: Continuous-array comparison on the common grid. D5 frame: $|\Delta V| \leq 1 \times 10^{-4}$ pu, $|\Delta f| \leq 1 \times 10^{-4}$ Hz, $|\Delta\theta| \leq 0.01°$, $|\Delta P|/|\Delta Q| \leq 1 \times 10^{-3}$ pu, $|\Delta I| \leq 1 \times 10^{-3}$ pu. Peak/settling $\leq 1\%$. Tier label: 'fill by generator'.

PENDING — table slot awaiting generator output (no measured numbers in this stub)

10.4 Counter comparison

Table 6: Counters as measured — reported, never gated.

PENDING — table slot awaiting generator output (no measured numbers in this stub)

10.5 Measured speedup — the headline

Table 7: **Measured speedup — the headline number of this report.** Same case, same machine, same schedule; B/A of the two engines' wall clocks. Timing row discloses (v0 scope — full event engine parity = v2, deferred, not cut); the machine-load variance warning (Machine-load note in the reference record §9) applies to every cross-session ratio shown for context only.

PENDING — table slot awaiting generator output (no measured numbers in this stub)

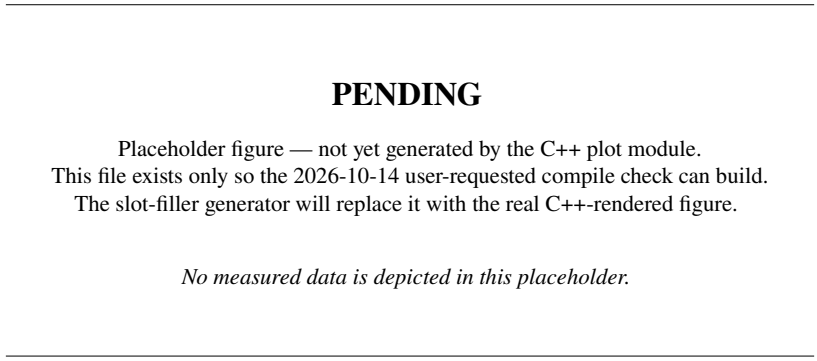


Figure 3: Top strips: per-IBR GFM/GFL mode of the four-mode enumerator and the SG31 breaker state (per the reference layout); (a) per-device active power; (b) per-device reactive power — $t \in [0, 160]$ s, full window, raw C++ accepted samples (every sample plotted; no smoothing, filtering, decimation, clipping, offset, or interpolation), device colors as bound in the preamble of this skeleton.

[PENDING: run-id + manifest sha256 citation to be generated here.]

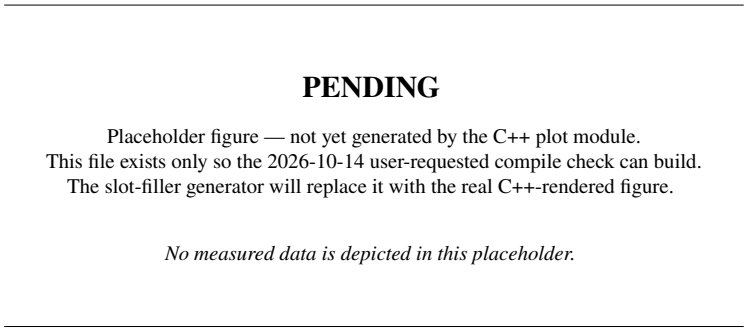


Figure 4: Eight-panel electrical response of the C++ engine — layout, colors, and legend placement cloned from the reference `matlab\_normal\_electrical.pdf`. Every plotted sample is a raw accepted value; nothing is smoothed, filtered, decimated, clipped, offset, interpolated, or augmented with synthetic noise.

[PENDING: run-id + manifest sha256 citation to be generated here.]

10.6 SG31 swing dynamics — C++ vs golden reference

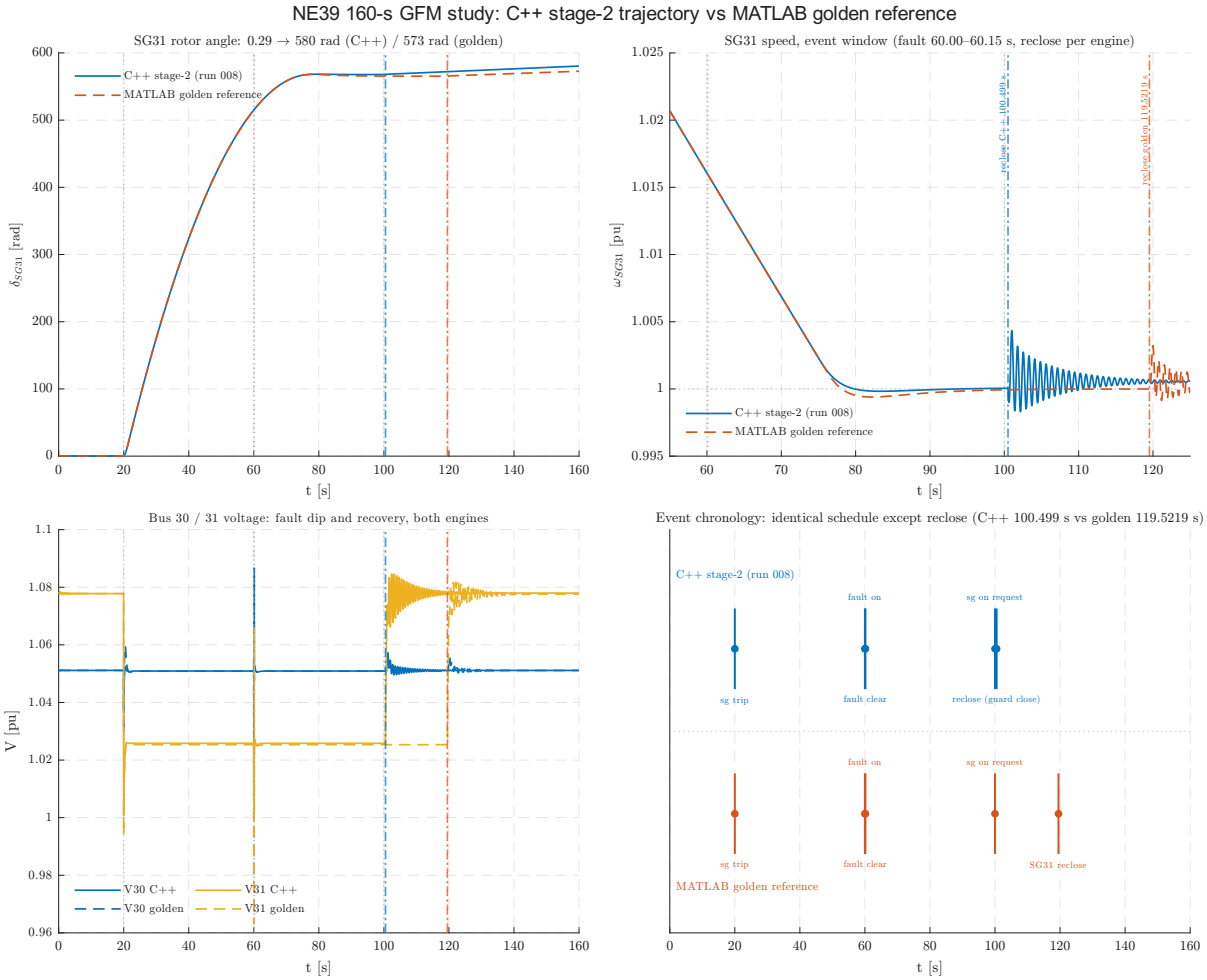


Figure 5: SG31 swing comparison — C++ stage-2 run `stage2\_smoke\_ne39\_008\_gfm160s\_golden` against the MATLAB golden reference. Top left: rotor angle δ_{SG31}, 0.29 → 580.4 rad (C++) / 572.8 rad (golden); top right: speed ω_{SG31} over the event window (55–125 s), showing the first swing 1.02 → 1.00 p.u.and the post-reclose oscillation of each engine; bottom left: bus 30/31 voltage with the 60.00–60.15 sfault dip and the reclose recovery; bottom right: event chronology. Both reclose instants are marked on every panel — C++ guard close 100.4990 sand golden 119.5219 s. Curves are raw samples with uniform 1:40 stride decimation for vector size only; no smoothing, filtering, or interpolation is applied.

The 19.02 sreclose gap is a *policy* gap — *when* each engine is permitted to close — not a trajectory defect. Numbers computed from the plotted series: maximum speed deviation at $t \geq 130$ sis 3.91×10^{-4} p.u.; bus 30 / bus 31 voltage deviation at $t \geq 120$ sis 5.17×10^{-3} / 7.62×10^{-3} p.u.. The rotor-angle drift rate at $t \geq 120$ sis 0.206 rad/s (C++) against 0.176 rad/s (golden); for an islanded machine that never settles to a fixed δ, the drift rate is the honest comparison metric and the accumulated angle is not.

11 Limitations and interpretation

- **v0 scope — full event engine parity = v2 (deferred, not cut).** The C++ engine in this report runs in fixed-step schedule-replay mode (C++ = fixed-step schedule-replay (v0) | MATLAB = adaptive strict-LTE). The adaptive strict-LTE stepping, the full event-engine transaction logic, and the private-trial machinery of the MATLAB reference are not part of the v0 claim; the v2 scope carries them.
- **Machine-load variance is the standing caveat of every cross-session ratio.** Same-day A/B-style pairs on an idle machine are the only directly comparable timings (the reference record's §9 methodology note carries the 3–4× morning-session evidence). Cross-session ratios in this report are contextual only.
- **Counter conventions are frozen.** Rejected steps are always attempts minus accepted on the public accepted path; the `progress.log` `rejects=` column is never used; the `jacobian\_evaluations` convention is unchanged.
- **The historical 2026-10-11 baseline cannot be decomposed** into early-phase vs remainder (no per-line timestamps in its `progress.log`); unsigned smoke-horizon ratios are never used.
- **No near-success phrasing.** Where a measured result is unwelcome (slow, overhead-dominated, or delayed), this report states it with the number and the identified follow-up work — never as "nearly successful" or "promising".
- **Criteria evolution is traceable.** The MATLAB reference record was accepted under D1–D5 (D5 Tier 2 DECISION-exact); the user's later ruling relaxed DECISION equality to "different + physical reason in the divergence ledger" with reclose SUCCESS as the only absolute gate. Both frames are documented; every comparison in this report states which frame it uses.
- **Counter conventions and raw-sample contract.** Every plotted sample in the figures is a raw accepted value of the C++ run record; nothing is smoothed, filtered, decimated, clipped, offset, interpolated, or augmented with synthetic noise. Douglas-Peucker simplification exists only in GUI previews, never in report figures.

12 Conclusion

[PENDING — slot not yet filled. Placeholder stub created 2026-10-14 for the user-requested compile check; the C++ report generator (slot-filler) will overwrite this file. No measured numbers are shown in this stub.]

13 Provenance table

Table 8: Source of every number in this report. C++ claims cite their run folder and manifest sha256 (cache convention); MATLAB claims cite the reference record.

PENDING — table slot awaiting generator output (no measured numbers in this stub)

References

[1] `docs/source/report_port_engine_en.pdf` — MATLAB reference record (same-tree A/B measurement, 2026-10-14) that this flagship report verifies against.

[2] `GridFormLab-port/docs/decisions_20261014.md` — D1–D5 acceptance rulings (authoritative).

[3] `docs/project/baseline_ne39_reclose_160s.md` — historical baseline reference (informational anchor).

[4] User run-cache directive 2026-10-14 — every C++ run persists case/request snapshot + raw trajectory + counters/DECISION + summary + manifest sha256, with automatic canonical dump; run history is the provenance source of this report.